Zhuoli Yin

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Research Interests

AI/ML-Augmented Combinatorial Optimization, Reinforcement Learning, Transportation and Logistics Systems, Sustainable Computing

Academic Employment

- **University of South Carolina** *08/2026 - present*
  Assistant Professor Columbia, SC, USA
  Department of Mechanical Engineering

Education

- **Purdue University** *08/2021 - 08/2026*
  Ph.D. in Industrial Engineering, *Advisor: Dr. Hua Cai* West Lafayette, IN, USA
  - Interdisciplinary Computational Science and Engineering Graduate Program
  - Graduate Certificate in Applied Statistics

- **Purdue University** *08/2019 - 08/2021*
  M.S. in Industrial Engineering, *Advisor: Dr. Hua Cai* West Lafayette, IN, USA

- **Beihang University** *09/2015 - 07/2019*
  B.E. in Electronic and Information Engineering Beijing, China

- **Polytechnic University of Madrid** *02/2019 - 05/2019*
  Exchange undergraduate student in Computer Science Madrid, Spain

Industry Experience

- **Amazon** *06/2025 - 09/2025*
  Applied Scientist Intern, *Global Transportation Services* New York, NY, USA
  - Developed Amazon's next-generation AI-driven decision-making tools for resilient and dynamic middle-mile network design.

Publications

Published Google Scholar

[J1] Khan, U., Guo, X., & **Yin, Z.** (2026). Worker Responses to Automated Safety Warning Systems in Construction Work Environments: A Review with AI-Assisted Screening. *Infrastructures, 11(11), 408.* https://doi.org/10.3390/infrastructures11110408

[C2] **Yin, Z.**, Ding, Y., Khir, R., & Cai, H. (2026). ViTSP: A Vision Language Models Guided Framework for Solving Large-Scale Traveling Salesman Problems. *The 14th International Conference on Learning Representations (ICLR).* https://proceedings.iclr.cc/paper_files/paper/2026/file/d6e71be5a6cb61dfa3f9ffbdd6e05b3c-Paper-Conference.pdf

[J3] **Yin, Z.**, Karakaya, E., Bass, K., & Cai, H. (2026). Arthur: An artificial intelligence powered teaching assistant system for Engineering Economics class. *International Journal of Artificial Intelligence in Education, 36(1), 100003.* https://doi.org/10.1016/j.ijaied.2026.100003

[J4] Liu, Y., **Yin, Z.**, & Cai, H. (2025). Enhanced global oil spill dataset from 1967 to 2023 based on text-form incident information. *Scientific Data, 12, 1394.* https://doi.org/10.1038/s41597-025-05601-9

[C5] **Yin, Z.**, Kou, Z., & Cai, H. (2023). A Deep Reinforcement Learning Model for Large-Scale Dynamic Bike Share Rebalancing with Spatial-Temporal Context. *In Proceedings of the 12th International Workshop on Urban Computing. ACM, Long Beach.* https://urban-computing.com/urbcomp2023/file/UrbComp2023_paper_7.pdf

[J6] **Yin, Z.**, Hardaway, K., Feng, Y., Kou, Z., & Cai, H. (2023). Understanding the Demand Predictability of Bike Share Systems: A Station-Level Analysis. *Frontiers of Engineering Management, 10(4), 551–565.* https://doi.org/10.1007/s42524-023-0279-8

Technical Reports

[R7] Tumula, R. S., **Yin, Z.**, Cai, H., & Gkritza, N. (2026). Development of Indiana-Specific Trip Generation Procedures and Models of E-commerce Demand: Implications for INDOT's Travel Demand Modeling and Investment Planning. *Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2026/25 (SPR-4906)*. West Lafayette, IN: Purdue University. https://doi.org/10.5703/1288284318707

PRESENTATIONS

[1] **Yin, Z.**, Khir, R., & Cai, H. (2026). PACER: Learning Time-Budget-Aware Pacing for Neural Local Search. *Oral Presentation at the 2026 INFORMS Annual Meeting*, San Francisco, CA, USA.

[2] Sustainable Development Track Special Session on "Does AI Belong at the Sustainability Table?" *Panelist, 2026 IISE Annual Conference*, Arlington, TX, USA

[3] **Yin, Z.**, Ding, Y., Khir, R., & Cai, H. (2026). When Does AI Use Help Improve Sustainability? A Marginal Carbon Accounting Framework for AI-enabled Vehicle Routing. *Oral Presentation at the 2026 IISE Annual Conference*, Arlington, TX, USA.

[4] **Yin, Z.**, Ding, Y., Khir, R., & Cai, H. (2026). ViTSP: A Vision Language Models Guided Framework for Large-Scale Traveling Salesman Problems. *Oral Presentation at the 2026 TRB Annual Meeting*, Washington, DC, USA.

[5] **Yin, Z.**, Ding, Y., Khir, R., & Cai, H. (2025). Rethink Artificial Intelligence for Vehicle Routing: Towards Generalization and Sustainability. *Oral Presentation at the 2025 INFORMS Annual Meeting, Job Market Showcase Track*, Atlanta, GA, USA.

[6] **Yin, Z.**, Ding, Y., Khir, R., & Cai, H. (2025). ViTSP: A Vision Language Models Guided Framework for Large-Scale Traveling Salesman Problems. *Oral Presentation at the 2025 IISE Doctoral Colloquium 3MT Competition*, Atlanta, GA, USA.

[7] **Yin, Z.**, Karakaya, E., Bass, K., & Cai, H. (2025). Using Artificial Intelligence Powered Teaching Assistant Tool to Enhance Student Learning in Engineering Economics Class. *Oral Presentation at the 2025 IISE Annual Conference*, Atlanta, GA, USA.

[8] **Yin, Z.**, Khir, R., & Cai, H. (2025). Integrated Dynamic Rebalancing and Charging Strategy for Electric Scooter Sharing System. *Oral Presentation at the 2025 IISE Annual Conference*, Atlanta, GA, USA.

[9] **Yin, Z.**, Ding, Y., Khir, R., & Cai, H. (2025). ViTSP: A Vision Language Models Guided Framework for Large-Scale Traveling Salesman Problems. *Oral Presentation at the 2025 IISE Annual Conference*, Atlanta, GA, USA.

[10] **Yin, Z.**, Khir, R., & Cai, H. (2024). Integrated Dynamic Rebalancing and Charging Strategy for Electric Scooter Sharing System. *Oral Presentation at the 2024 INFORMS Annual Meeting*, Seattle, WA, USA.

[11] **Yin, Z.**, & Cai, H. (2024). DeepBike: A Deep Reinforcement Learning Based Model for Large-scale Online Bike Share Rebalancing. *Poster Presentation at the TRB Annual Meeting 2024*, Washington, DC, USA.

[12] **Yin, Z.**, & Cai, H. (2023). Optimal Rebalancing and Charging Strategies for Life Cycle Net Emissions for Shared Electric Scooter System. *Oral Presentation at the 2023 INFORMS Annual Meeting*, Phoenix, AZ, USA.

[13] **Yin, Z.**, Kou, Z., & Cai, H. (2023). DeepBike: A Deep Reinforcement Learning Based Model for Large-scale Online Bike Share Rebalancing. *Oral Presentation at the 2023 Purdue Research Symposium on Operations*, West Lafayette, IN, USA.

[14] **Yin, Z.**, Kou, Z., & Cai, H. (2023). DeepBike: A Deep Reinforcement Learning Based Model for Large-scale Online Bike Share Rebalancing. *Oral Presentation at the 12th International Workshop on Urban Computing (with KDD 2023)*, Long Beach, CA, USA.

[15] Liu, T., **Yin, Z.**, Hua, I., & Cai, H. (2023). Using Data Linkage Approach to Refine the Industrial Water Consumption Data Resolution in the United States. *Poster Presentation at the Ecological Sciences and Engineering 2023 Symposium*, West Lafayette, IN, USA.

[16] **Yin, Z.**, Kou, Z., & Cai, H. (2022). DeepBike: A Deep Reinforcement Learning Based Model for Large-scale Online Bike Share Rebalancing. *Oral Presentation at the 2022 INFORMS Annual Meeting*, Indianapolis, IN, USA.

[17] **Yin, Z.**, Kou, Z., & Cai, H. (2022). DeepBike: A Deep Reinforcement Learning Based Model for Large-scale Online Bike Share Rebalancing. *Oral Presentation at the 2022 IISE Annual Conference*, Seattle, WA, USA.

Research Projects and Collaborations

- **Graduate Research Assistant** *2025*
  Development of Indiana-Specific Trip Generation Procedures & Models of e-Commerce Demand: Implications for INDOT's Travel Demand Modeling and Investment Planning
  - Funder: Indiana Department of Transportation (INDOT)
  - PIs: Dr. Nadia Gkritza and Dr. Hua Cai
  - Responsibility: Co-lead the project and develop agent-based models integrating e-commerce demand with daily personal travel patterns based on survey data in Indianapolis.

- **Graduate Research Assistant** *2022*
  Teaching Innovation Project: Developing an AI-powered "TA" tool for IE343 (Engineering Economics)
  - Funder: Purdue Innovation Hub
  - PIs: Dr. Hua Cai and Dr. Erhan Karakaya
  - Leveraged Gradescope data to develop a web-based intelligent educational tool that predicts causes of incorrect answers and delivers personalized feedback.
  - Collaborated with software engineers and instructors for classroom deployment and evaluated the tool's effectiveness, resulting in one published paper (IJAIED, 2026) and one paper under review.
  - Highlighted by Purdue Today Article (A new kind of teaching assistant), and Purdue InnovatED Graduate Research Magazine (Meet Arthur: Your AI Virtual Teaching Assistant).

- **Graduate Research Assistant** *2021*
  A Computational Approach for Industrial Water Consumption Estimation, Prediction, and Impact Evaluation

- Funder: Purdue College of Engineering Initiative on Smart City
- PIs: Dr. Inez Hua and Dr. Hua Cai
- Responsibility: Co-Led the project and developed optimization model and natural language processing approach for pairing large-scale water withdrawal and discharge.

Teaching Experience

- **Teaching Interests**
  Engineering Economics, Linear Programming, Integer Programming, Dynamic Programming, Stochastic Models, Probability and Statistics, Manufacturing Processes, Supply Chain and Logistics Modeling, Systems Simulation

- **Teaching Certificates** *Fall 2022*
  Certificate in Foundations of College Teaching, Purdue University

- **Instructor, Purdue University** *Fall 2023*
  IE 343 - Engineering Economics (In-person)
  - Enrollment: 148 undergraduate students from 10 different engineering disciplines.
  - Student evaluation score: 4.5 out of 5.0
  - Fully responsible for the class, and supervised two teaching assistants and two graders.
  - Topics covering the time value of money, cash flow analysis, capital depreciation, and cost-benefit analysis.
  - Administered a flipped class model to promote student-centered active learning.
  - Maintained an AI-based virtual teaching assistant tutoring system, supporting the practice of assignments.
  - Evaluated students' learning outcomes for Accreditation Board for Engineering and Technology (**ABET**)

- **Guest Lecturer, Purdue University** *Fall 2024*
  EEE 560 - Environmental Data Science (In-person)
  - Enrollment: 37 graduate students
  - Developed a session on leveraging artificial intelligence for environmental problem solving using applicable data.
  - Led the discussion on the state-of-the-art artificial intelligence algorithms.
  - Provided feedback to students' group research proposals.

- **Teaching Assistant, Purdue University** *Fall 2021 - Fall 2025*
  IE 343 - Engineering Economics (In-person; Online in Spring 2022 and Summer 2024)
  - Nine sections in total
  - Enrollment: 100-200 undergraduate students per section
  - Average student evaluation scores across eight sections: 4.6 out of 5.0
  - Hosted regular office hours and led the online forum discussion.
  - Mentored graders for developing solution keys and grading assignments.
  - Taught review sessions to facilitate the preparation for the exams.
  - Designed exams and case study projects.
  - Provided ABET assessment materials for the courses.

- **Teaching Assistant, Purdue University** *Spring 2024*
  IE 590 / EEE 595 - Urban Mobility Optimization (In-person)

◦ Enrollment: 22 graduate students

◦ Student evaluation score: 4.8 out of 5.0

◦ Developed example codes to showcase the application of Gurobi modeling.

◦ Organized the forum discussion on the latest topics in urban mobility, such as electric vehicles, autonomous vehicles, and ride sharing.

◦ Mentored the case study project and supported coding, data processing, and results analysis.

Research Advising

University of South Carolina

PhD Students

Fall 2026– Usama Khan, Texas State University (Co-advised with Dr. Xingzhou Guo)

Senior Design Mentor

Fall 2026– Project Title: **Data Center Direct Liquid Cooling System**
Students: Martin Dwyer, Tommy Dinan, James Duyan, George Lane, and Reid Vaughn

Purdue University

Undergraduate Students

08/2023–12/2023 Quan The Dinh (CS), Purdue University

02/2023–05/2023 Karen Fortunat (IE), Purdue University

02/2023–05/2023 Elaine Sui (IE), Purdue University

08/2022–02/2023 Laura Almeida Tinjaca, National University of Colombia–Purdue Program (Admitted to the MS program in Environmental and Ecological Engineering at Purdue University with Lynn Fellowship)

05/2022–08/2022 Kalei Bass (IE), Purdue University

Professional Service

• **Journal Reviewer**

◦ *Journal of Cleaner Production*

◦ *Engineering Applications of Artificial Intelligence*

◦ *Networks and Spatial Economics*

◦ *IEEE Transactions on Intelligent Transportation Systems (ITS)*

◦ *Transportation Research Part D: Transport and Environment*

◦ *Journal of Simulation*

• **Conference Reviewer**

◦ Neural Information Processing Systems (NeurIPS) 2025, 2026

◦ Transportation Research Board Annual Meeting (TRBAM) 2026, 2027

• **External Service to Professional Societies**

◦ Session Chair, IISE Annual Conference & Expo 2022

◦ Support team, 2023 Purdue Research Symposium on Operations

• **Internal Service at Purdue University**

◦ Vice President, Industrial Engineering Graduate Student Organization (05/2022 - 05/2023)

- Organized the Industrial Engineering research symposium.

- Coordinated a series of bi-weekly IE graduate student research showcases.
- Invited external speakers and facilitated workshops on Resume writing

◦ Judge, 2022 Summer Undergraduate Research Symposium, College of Engineering (07/2022)

◦ Graduate Student Representative, Industrial Engineering Graduate Showcase (08/2022)

◦ Moderator, Prospective Ph.D. Student Panel, School of Industrial Engineering (03/2023)

◦ Student representative on the faculty search committee (10/2023 - 12/2023)

Honors and Awards

- **Estus H. and Vashti L. Magoon Research Excellence Award**, College of Engineering, Purdue University, 2026
- **IISE Doctoral Colloquium** (3 participants nominated per department), 2025
- **NSF IISE Annual Conference Student Travel Award**, 2025
- **IISE Future Faculty Fellow** (15 participants selected per year), 2024–2025
- **InnovatED Graduate Research Magazine**, Purdue University (10 out of 65 articles selected), 2024
- **Lee A. Chaden Fellowship in Industrial Engineering**, Purdue University, 2023–2024
- **Graduate Student Government Travel Award (Tier 1)**, Purdue University, 2023
- **Graduate School Summer Research Grant**, Purdue University, 2023
- **Frederick N. Andrews Environmental Travel Grant Award**, Purdue University, 2023
- **Student Scholarship**, Seth Bonder Foundation INFORMS, 2022
- **Honorable Mention Poster Award** (5.3% award rate), Purdue Engineering Graduate Showcase, 2022
- **Center for the Environment Travel Grant**, Purdue University, 2022